Total No. of Questions : 10]

SEAT No. :

PA-346

[Total No. of Pages : 2

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B.E. (Mechanical Engineering)

ADVANCED MANUFACTURING PROCESSES

(2015 Pattern) (Semester - II) (Elective - IV) (402050 A)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *All questions are compulsory i.e. Solve Que. 1 or Que. 2, Que. 3 or Que. 4, Que.5 or Que.6, Que.7 or Que.8 and Que.9 or Que.10.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*

Q1) a) Examine with sketch electro hydraulic forming. **[6]**

b) How friction stir welding is useful in modern era? **[4]**

OR

Q2) a) Examine with neat sketch Stretch forming and list their applications. **[6]**

b) List applications of adhesive bonding. **[4]**

Q3) a) Examine with neat sketch Electro Jet Machining process with its process parameter. **[6]**

b) Examine the process of underwater welding. **[4]**

OR

Q4) a) Examine working principle of wire electric discharge machining with the process parameter. **[6]**

b) Examine basic principle of friction stir welding process. **[4]**

Q5) a) Examine the process of Focused Ion Beam Machining. **[6]**

b) Examine how- the ultrasonic micro machining carried out. **[6]**

c) Write short note on Diamond micro machining. **[4]**

OR

P.T.O.

Q6) a) Write in detail the need of micro machining. **[6]**

b) Examine the process of photochemical machining. **[6]**

c) Write short note on Lithography. **[4]**

Q7) a) Give classification of additive manufacturing processes. **[6]**

b) Write note on post processing of parts manufactured by additive manufacturing processes. **[6]**

c) Write applications of additive manufacturing processes in medical technology. **[4]**

OR

Q8) a) Examine the basic steps in additive manufacturing. **[6]**

b) What care should be taken while designing the object while manufacturing by additive manufacturing. **[6]**

c) Write application of additive manufacturing processes in aerospace industry. **[4]**

Q9) a) Examine operating principal of Atomic force micro scope with neat sketch. **[6]**

b) Examine operating principal of Electron Microscopes with neat sketch. **[6]**

c) Describe in details, the applications of microscopes. **[6]**

OR

Q10) a) Examine operating principal of Energy-dispersive X-ray spectroscopy. **[6]**

b) Describe in details, the applications of spectroscope. **[6]**

c) Write short note on material characterization. **[6]**